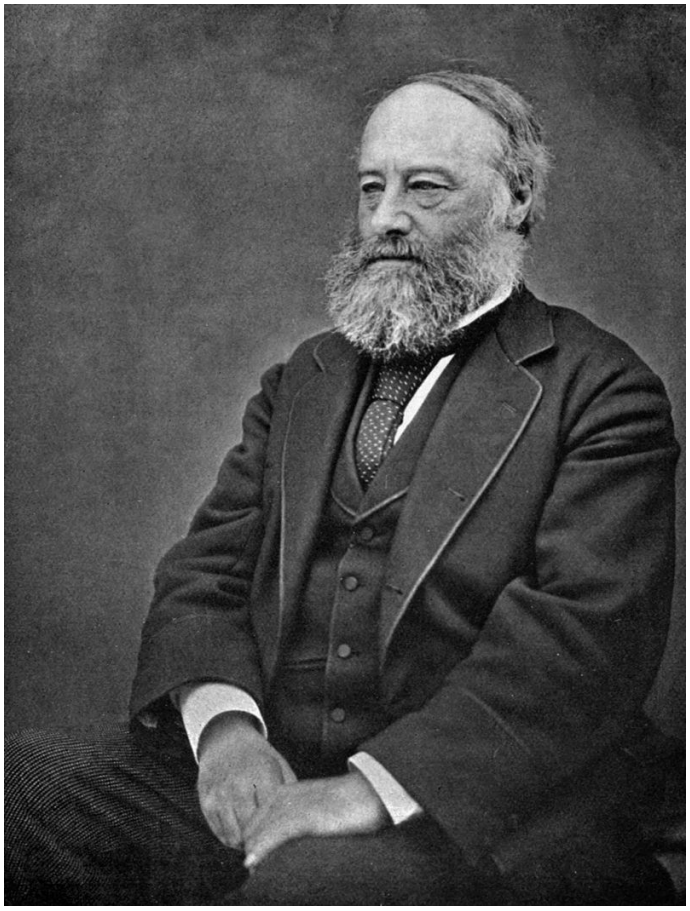
普通物理實驗結報



熱功當量

第三組

組員:	貢獻度
吳	35%
卓	35%
程	30%

1 實驗目的

To find the heat-work equivalent (熱功當量) by using the electrothermal method, and to find the system's relation to heat dissipation.

2 實驗原理

The Mechanical Equivalent of Heat (熱功當量) is a *constant* which determines how much mechanical work is equal to a certain amount of heat. It is defined as:

$$W = J\Delta H$$

Note:

- W : Mechanical work ($Joule$).
- ΔH : Heat ($Calorie$).
- J : The heat-work conversion constant / 熱功當量 (J/cal).

The formula is derived from the **principle of energy conservation**, which in this scenario can be interpreted as: *"For every external work W, it will be completely converted to internal energy (ΔH)"*. But, this derivation assumes an **ideal system** in which all external work is conserved within the system and fully transformed into internal energy (ΔH), with no heat loss or other dissipative effects.

Since in this experiment, our system is not **perfectly isolated**, some of the external work will not contribute to increasing the system's internal energy as heat. Instead, part of it will be lost to the environment due to *heat dissipation*. Therefore, the actual change in internal energy (ΔH) is *slightly less* than the total work applied, with the difference primarily accounted for by heat loss.

Before discussing heat dissipation, it is important to clarify what we mean by heat. Simply put, **heat is a measure of the kinetic energy of the particles within a substance**. Temperature is the observable result of this particle motion, reflecting the average kinetic energy. The quantitative relationship between heat and temperature

is described by the formula:

$$Q = mc\Delta T$$

where Q (ΔH) is the heat transferred, m is the mass of the substance (Joule), c is the specific *heat capacity* ($J/(kg°C)$, and ΔT is the *temperature change* ($°C$).

For a **group of particles**, the change in internal energy can be expressed macroscopically as:

$$\Delta H = C\Delta T$$

where $C = \sum_{i=1}^{N} m_i c_i$ is the system's *total heat capacity*.

Next, since the system is *not perfectly isolated* (unlike in the idealized case), there is bound to be a mismatch between the *theoretical* maximum temperature and the temperature *measured* in the field. As previously mentioned, this difference is mainly due to **heat dissipation**, which can be described by **Newton ' s Law of Cooling**. From this, we can conclude that:

$$T_{measured} = T_{theoretical} - \Delta T$$

Newton's Law of Cooling

Newton's Law of Cooling is a principle that describes how a particle ' s thermal energy dissipates due to its surroundings. It can be summarized as:

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*"The rate at which an **object** loses heat is **proportiona**l to the difference between its temperature and that of its surrounding environment."*

Mathematically, it is defined as:

$$u = \frac{dT}{dt} = \kappa(T - \theta)$$

- where u is the temperature's *rate of change*.
- T , in this formula, is defined as the *the system's temperature* with θ as the *external temperature.*
- κ is the **cooling constant** of the system.

And through algebra and calculus, we can **derive** κ as:

$$\kappa = -\Delta t \ \ln(\frac{T_r - \theta}{T_f - \theta})$$

Note:

- The reason κ is negative is that it represents **cooling**. Since the temperature of the object decreases over time, the *rate of change of temperature is negative*. Using a negative value for κ **correctly** reflects this decrease in temperature according to the convention of Newton ' s Law of Cooling.
- T_r represents the **initial temperature**, and T_f represents the **final temperature**.

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3 實驗方法

實驗依據能量守恆原理，將系統總熱容量 C 的測定與電能轉換的測量相結合，並搭配散熱修正機制：

- **熱容量測定**：利用混合法（Method of Mixtures）測量系統（銅杯、攪拌器）的有效熱容量，再與水的熱容量相加，得到實驗所需的總熱容量 C。
- **電能轉換**：透過量測已知的電壓 V、電流 I 及加熱時間 Δt，計算輸入系統的總電能 $W = VI\Delta t$。
- **能量守恆關係**：將輸入的電能 W 視爲與系統熱能增加 $C\Delta T$ 成正比，即 $W = J \cdot (C\Delta T_{correct})$。
- **散熱修正機制**：
  - 測量加熱結束後的降溫曲線（Cooling Curve）。
  - 利用降溫曲線依據牛頓冷卻定律 $\frac{dT}{dt} = \kappa(T - \theta)$，求得散熱係數 κ。
  - 將 κ 應用於升溫過程，計算出因散熱損失的溫差 dT，從而將實測溫升 $\Delta T_{measured}$ 修正爲 $\Delta T_{correct} = \Delta T_{measured} + dT$，確保結果的準確性。

4 數據分析

4.1 實驗一: 測定卡計熱容

- 溫度較高水質量 M=150.15(g)，溫度 T_H =40.2($^\circ C$)
- 溫度較低水質量 M=150.17(g)，溫度 T_0 =25.2($^\circ C$)
- 混合後系統溫度 T_{ave} =33.6($^\circ C$)

$$\text{系統熱容 } C = \frac{Sm(T_{ave} - T_0)}{T_H - T_{ave}} - SM = 40.98(cal/^\circ C)$$

4.2 實驗二: 熱功當量計算

- 銅杯內水質量 $m_{H_2O} = 250.03(\mathrm{g})$，溫度 $T_0 = 24.5(°\mathrm{C})$。
  室溫 (θ)=25.9(°C)，加熱電壓 (V) =9.99(V)，卡計電阻 (R) =16.2(Ω)，
  輸出電流 (A) =0.617(A)。

- 時間 t 與溫度 T 的關係圖

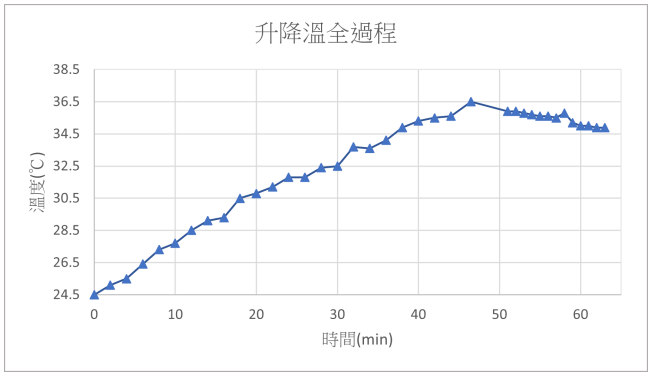

Figure 1: 時間 t 與溫度 T 的關係圖

- 加熱總時間:$t_f = 46.5(\mathrm{min})$，t_f 時刻量得的系統溫度 $T_f = 36.5(°\mathrm{C})$，
  停止加熱時間 $t_R = 16.5(\mathrm{min})$，t_R 時刻量得的系統溫度 $T_R = 34.9(°\mathrm{C})$，
  $\kappa = \frac{1}{t_R - t_f}\ln(\frac{T_R-\theta}{T_f-\theta}) \approx -0.0055$，散熱導致的溫差 $\Delta T = 1.17°\mathrm{C}$;
  修正後的末溫 $T_H = T_f + \Delta T = 36.5°\mathrm{C} + 1.17°\mathrm{C} = 37.67°\mathrm{C}$;

- 外界提供的總電能：總加熱時間 $t_f = 46.5\ \mathrm{min} = 2790\ \mathrm{s}$。

$$W = V \cdot I \cdot t_f = 9.99\ \mathrm{V} \cdot 0.617\ \mathrm{A} \cdot 2790\ \mathrm{s} \approx 17197.09\ \mathrm{J}$$

- 系統吸收的總熱能 Q 的計算 (假設系統總熱容量 $C_{total} = 290.98\frac{\mathrm{cal}}{°\mathrm{C}}$):
  修正後的溫升 $\Delta T_{correct} = T_H - T_0 = 37.67°\mathrm{C} - 24.5°\mathrm{C} = 13.17°\mathrm{C}$。

$$Q = C_{total} \cdot \Delta T_{correct} = 290.98\frac{\mathrm{cal}}{°\mathrm{C}} \cdot 13.17°\mathrm{C} \approx 3833.09\ \mathrm{cal}$$

- 熱功當量 J 的計算：

$$J = \frac{W}{Q} = \frac{17197.09\ \mathrm{J}}{3833.09\ \mathrm{cal}} \approx 4.487\mathrm{J/cal}$$

- 誤差分析：相較於公認值 4.186J/cal，本次實驗結果的誤差約爲 7.19%。

4.3 誤差分析 (Discussion on Errors)

根據實驗結果，測得的熱功當量 $J \approx 4.487\text{J/cal}$ 與公認值 4.186J/cal 相比，誤差約爲 **7.19%**。由於實驗值 J 偏高，根據 $J = W/Q$ 的關係，這意味著系統吸收的有效熱能 Q 被低估，或輸入電能 W 被高估。以下是推測的誤差來源。

4.3.1 系統性誤差

系統性誤差是導致 J 值偏高的主要原因，尤其與熱容量測定和散熱修正的準確性有關。

Table 1: 系統性誤差分析

來源	影響機制	最終影響
卡計熱容 C 測定不準確	實驗一混合法中，高溫水在混合前或混合過程中散失熱量，導致平衡溫度 T_{ave} 偏低，計算所得的卡計水質量 m 偏小。	C 偏小 $\rightarrow Q$ 低估 $\rightarrow J$ 偏高
電錶校準問題	若用於測量電壓 V 或電流 I 的電錶讀數存在系統性偏高，則導致計算所得的總電能 $W = VIt$ 被高估。	W 高估 $\rightarrow J$ 偏高
溫度計延遲/位置	溫度計的反應速度慢，未能及時捕捉到系統的實際最高溫 T_f，或溫度計未放置在水的平均溫度處。	ΔT 偏小 $\rightarrow Q$ 低估 $\rightarrow J$ 偏高

4.3.2 散熱修正的不足

散熱修正是本實驗精確性的關鍵。使用的單一 κ 值模型未能完全模擬眞實的熱傳遞過程。

- **冷卻係數非定值 κ**：散熱係數 κ 是通過降溫過程的數據計算得到的，但此計算假設 κ 在整個升溫過程中保持不變。在加熱過程中，系統溫度較高，熱輻射的影響（正比於絕對溫度的四次方 T^4）會更爲顯著，導致實際 κ 隨溫度升高而略微增加。採用單一 κ 值會 ** 低估 ** 高溫階段的散熱量。

- **修正方法近似：**無論是使用積分近似法或平均溫差法來計算總修正量 $\Delta T_{loss} = 1.17°\mathrm{C}$，都是基於牛頓冷卻定律的近似。如果升溫過程的溫度變化是非線性的，會導致修正量本身不夠精確，從而使得修正後的末溫 T_H 仍有偏差。

4.3.3 隨機誤差

- **計時與同步誤差：**在啓動/停止電源與碼錶計時之間不同步，會影響總加熱時間 t_f 的精確度。

5 結果與討論

In this experiment, the **mechanical equivalent of heat** was determined by combining *calorimetric* measurements with *electrical energy* input, under the assumption of energy conservation. The system heat capacity was first obtained using the **method of mixtures,** and the electrical work supplied during heating was calculated from ***measured voltage, current, and heating time.*** To improve accuracy, a heat-loss correction was applied using the cooling curve and Newton ' s law of cooling.

After applying the **heat-loss correction**, the corrected final temperature of the system was $T_H = 37.67\,^{\square}C$, corresponding to a corrected temperature rise of $\Delta T_{correct} = 13.17\,°C$ .Using a total system heat capacity of $C_{total} = 290.98\ cal/°C$ the absorbed thermal energy was calculated to be $Q \approx 3833.09\ cal$. The total electrical energy supplied was $W \approx 1.72{\times}104\ J$, leading to an experimentally determined *mechanical equivalent of heat,*

$$J = \frac{W}{Q} \approx 4.487\ J/cal.$$

This result is **higher** than the accepted value of $4.186\ J/cal$ by approximately 7.19%. Since $J = W/Q$, a higher experimental value suggests that the *absorbed heat* Q was **underestimated** or that the supplied *electrical energy* W was **overestimated**. Several factors likely contributed to this deviation. Systematic errors, such as an underestimated calorimeter heat capacity due to **heat loss** during the mixing process, would reduce the calculated Q. In addition, limitations in electrical measurement accuracy could lead to a **slight overestimation** of W. Temperature measurement effects, including thermometer response delay and non-uniform temperature distribution within the water, may also have caused the measured temperature rise to be smaller than the true average value.

The heat-loss correction, while necessary, introduces further approximation. The cooling coefficient κ was determined from the **post-heating cooling curve** under the assumption that it remains **constant** throughout the heating process. In reality, heat transfer mechanisms become *more significant at higher temperatures*, and the use of a single κ value may not fully account for heat loss during the **later stages of heating**.

Consequently, the corrected temperature rise may still be slightly underestimated, contributing to the observed overestimation of J.

Despite these limitations, the experimental data show a clear and consistent relationship between **electrical work** and **thermal energy increase.** The *magnitude* of the measured heat equivalent is reasonably close to the accepted value, and the observed error lies within an expected range for a calorimetry experiment with unavoidable heat exchange. Therefore, while the numerical accuracy is limited, the experiment can be considered an **overall success** in verifying the *principle of energy conservation* and demonstrating the practical challenges of precise heat measurements.

6 思考問題

Q_1 : T_0 **是否一定要是在系統加熱前的溫度? 可否任意選定? 是否加熱一開始即須計時?**

- T_0 在定義中是加入銅杯內水的溫度，並未與卡計熱平衡，因此 T_0 可以選用與卡計熱平衡後的溫度。
- 不可任意選定，因爲 T_0 影響卡計熱容的計算。
- 是。因爲計算熱功當量計算是以外界提供的電能來換算，所以不可任意更改時間，該行爲會造成電能計算不精確。

Q_2 : **以實驗數據說明散熱修正的必要性。並討論不進行修正會造成多大的誤差?**

- 在實驗過程中，即使停止加熱，水溫也會逐漸下降 (即降溫過程或冷卻曲線)。這段降溫本身就是系統持續向環境散熱的直接證據。故此修正的必要性可顯。
- 下表是修正散熱與不修正的比較:

	T	$T+\Delta T$	
溫度	36.50	37.67	
熱功當量	4.486	4.544	
誤差	7.2%	8.6%	熱功當量準確值:4.186

有此可得出，在不進行修正的前提，所測得的熱功當量相比修正後有更大的誤差 (7.2% → 8.6%)

Q_3 : **如何判定散熱修正後，末溫是正確的?**

- 如果修正後的熱功當量 J 值與公認值 (4.186 J/cal) 非常接近 (誤差小於 5%)，則可以推斷修正後的末溫 $T_{f_correct}$ 是正確的。

Q_4 : **分別利用以下兩式所得的溫差 ΔT 相差多少?**

$$\text{方法一: } dT = \int_t^{t+\Delta t} \kappa \cdot (T-\theta)\, dt \approx \Delta t \cdot [\kappa \cdot (T-\theta)]$$
$$\text{方法二: } |\Delta T = \kappa \cdot (\tfrac{T_f+T_0}{2} - \theta)|$$

1. 首先，使用 excel 列出每一段時間間隔的平均溫來計算 dT。
2. 結果如下表:

修正方法	方法一	方法二	
溫度	37.79	37.67	
熱功當量	4.446	4.544	
誤差	6.2%	8.6%	熱功當量準確值:4.186

可見方法一相比方法二準確度更高

Q_5 : **κ 值是否與溫度有關?**

- 是。散熱係數 κ 包含了所有熱傳遞的貢獻：傳導、對流和輻射。
- 傳導和對流：這兩種模式的熱傳遞係數對溫度有依賴性，但通常在實驗所涉及的溫差範圍內 (幾十攝氏度) 變化不大。
- 輻射：根據史蒂芬-波茲曼定律，輻射的散熱功率正比於 T^4。這表示輻射散熱對溫度非常敏感。當實驗溫度較高時 (例如超過 50°C)，輻射的影響會變得很顯著，此時 κ 就會明顯隨溫度升高而增大。